

SAFETYVERSE · BY UPL

The Really Simple Guide

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Live app: <https://upl-fire-portal.onrender.com>

Repository: <https://github.com/paramshah1903/fire-portal>

SafetyVerse — The Really Simple Guide

If you've never touched code in your life, this is for you. No jargon. No assumed knowledge. Just plain English and pictures.

1. What does this thing even do?

Think about how a **restaurant** works.

- Someone in the kitchen cooks the food.
- A waiter takes your order and brings your food.
- A cash register keeps track of every bill.

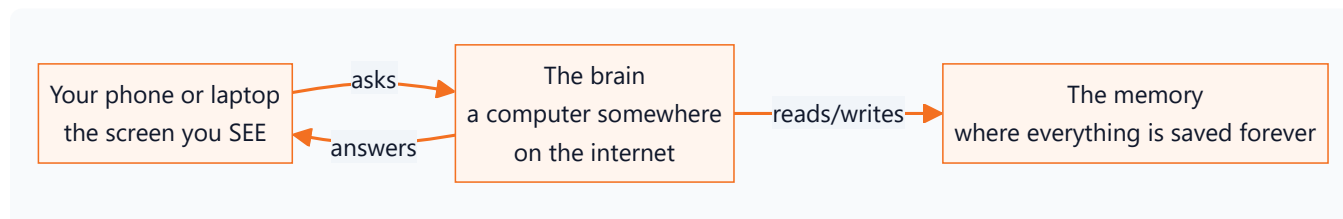
SafetyVerse is the same idea, but for fire extinguishers.

- Every fire extinguisher (or hydrant, or hose reel) gets a sticker with a **QR code** — like a name tag.
- Someone points their phone camera at the sticker.
- The phone opens a form asking "is this extinguisher OK?" — pass, fail, photo, done.
- The company's fire safety manager sees everything on one screen, from anywhere, live.

That's it. Everything else in this document explains the plumbing behind that simple idea.

2. The three parts

Any app you've ever used — WhatsApp, Netflix, your bank app — has **three parts**. Ours does too.

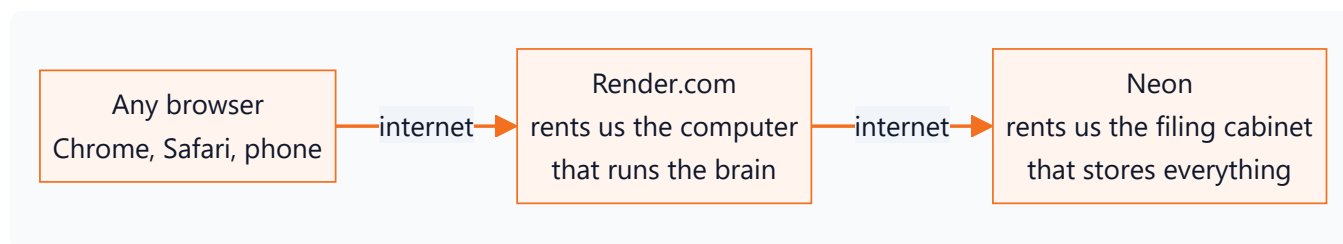


Part	Real-world analogy	In SafetyVerse
The screen	The waiter you talk to	The web page in your browser
The brain	The kitchen doing the work	A computer program running on rented servers
The memory	The restaurant's filing cabinet	A database that never forgets anything

Nothing more than that. Three parts talking to each other.

3. Where do the parts live?

We rent all three from other companies. We didn't buy any servers.



- **The screen** — lives on the same rented computer as the brain (they're bundled together)
- **The brain** — runs on **Render.com**. Free tier. It's like renting a small apartment for the app to live in.
- **The memory** — lives on **Neon**. Also free tier. A safe filing cabinet in the cloud.

Both companies charge zero dollars for what we're using right now. When SafetyVerse gets big enough to matter, we'll pay them a few dollars a month.

4. What lives in the filing cabinet?

The database is just **14 lists**. Each list is like a spreadsheet with a few columns.

Here are the important ones:

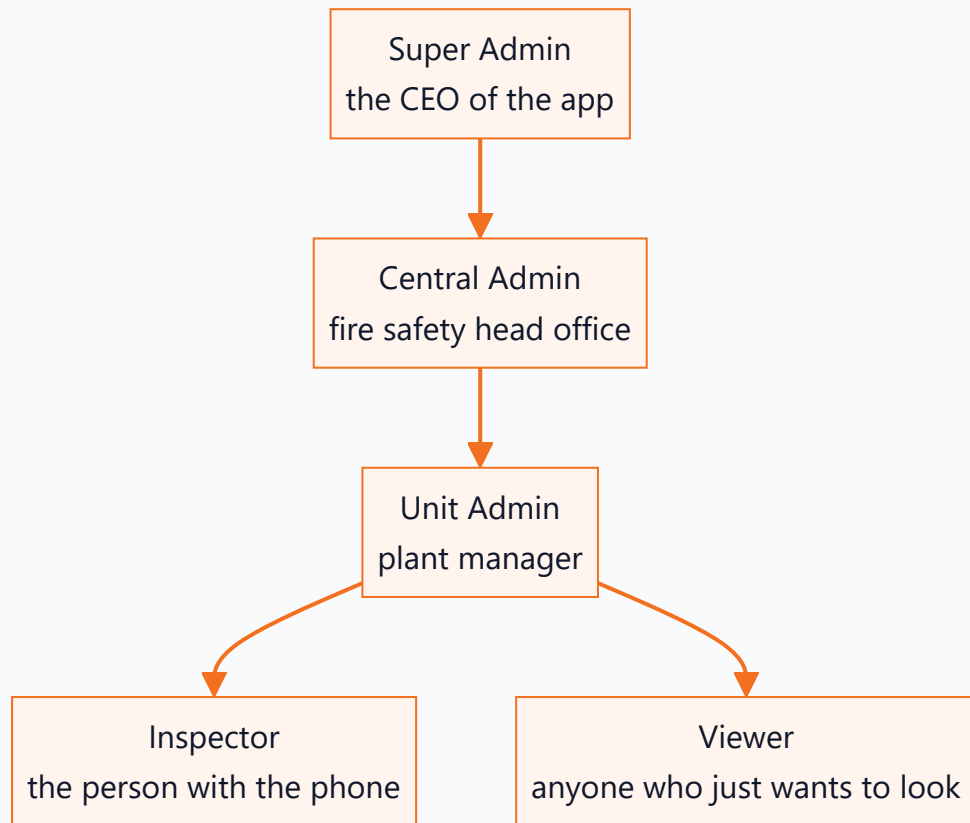
List	What's in it	Real-world equivalent
Users	Everyone who can log in	Employee directory
Roles	5 types: super admin, inspector, etc.	Job titles
Units	Every UPL plant location	Branch offices
Departments	Sub-areas inside a plant	Rooms in a branch
Equipment	Every fire extinguisher, hydrant, etc.	The physical stuff
Equipment Types	Categories (extinguisher, hydrant, ...)	Product catalogue
Checklist Templates	The questions to ask	Blank forms
Inspections	One record per equipment × month	Filled-in forms
Corrective Actions	Follow-ups when something fails	To-do list

The lists **connect to each other**. For example, every "inspection" knows which "equipment" it belongs to, which "person" did it, and which "unit" it happened at. So we can ask questions like *"show me every failed*

inspection at Plant 7 last month" and get an answer in a fraction of a second.

5. Who's allowed to do what?

Not everyone can do everything. There are **5 kinds of users**.

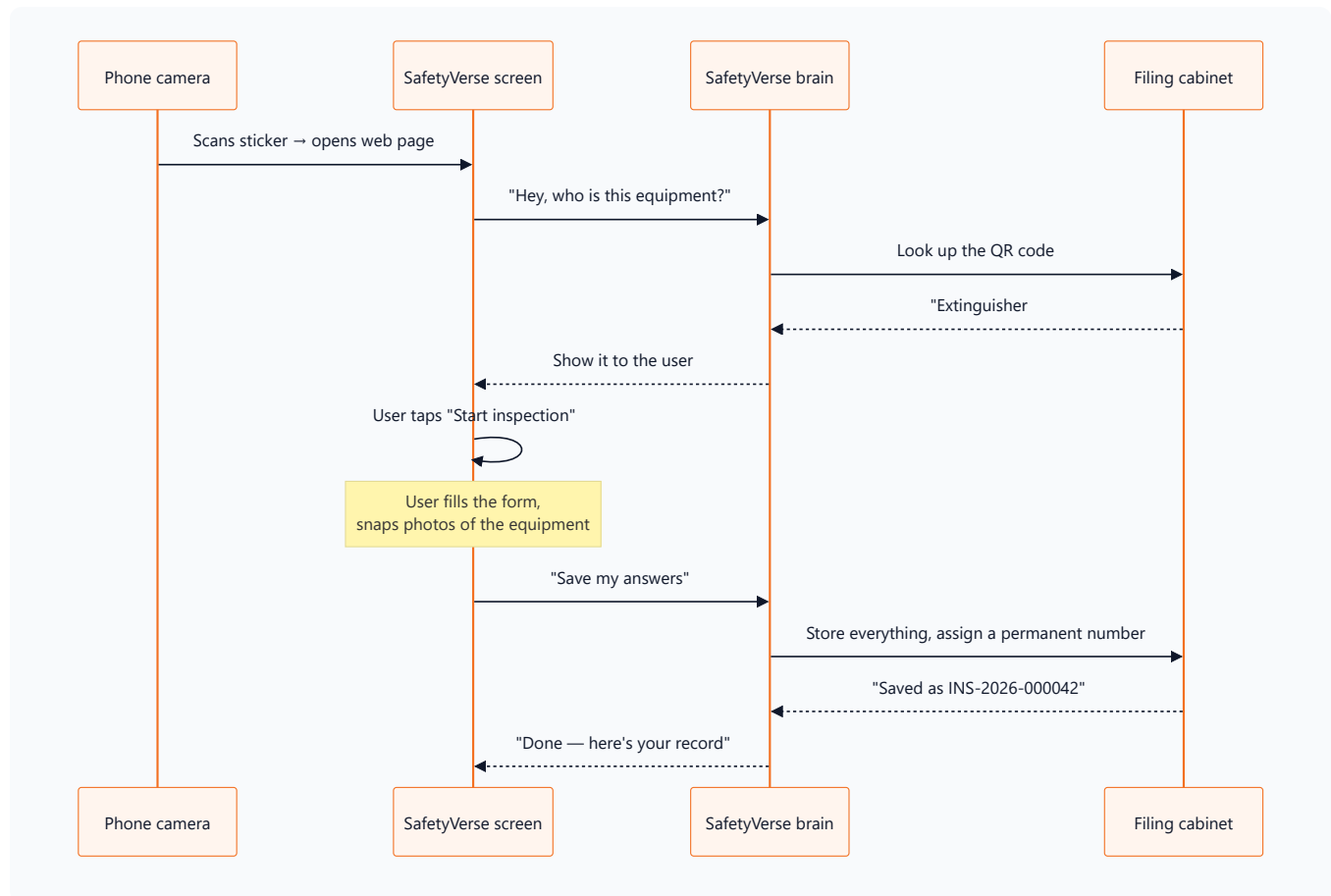


Role	What they can do	Analogy
Super Admin	Everything, everywhere	Owner of the restaurant
Central Admin	Manage forms + all inspections, but not create users	Head chef
Unit Admin	Everything inside their unit only	Restaurant manager
Inspector	Just do inspections in their unit	Waiter
Viewer	Read only. Can look, cannot change	Customer reading the menu

The app **hides menus** you're not allowed to use. And even if you try to sneak in through the back door, the brain rejects you. Two layers of safety.

6. What actually happens when you scan a QR code?

Let's follow one inspection from start to finish.



That's the whole flow. Six steps. Under 60 seconds if the inspector knows what they're doing.

If any answer was a "fail", the brain automatically creates a **follow-up task** so no one forgets to fix it. That's the killer feature.

7. What's inside the code repository?

The whole thing is just **two folders**.

```

upl_project/
├── frontend/  ← the screen the user sees
└── backend/   ← the brain
  
```

Inside each, code is organised into small folders by what it does. You don't need to know the details unless you're changing the code. If you're changing the code, read the [full developer guide](#).

One important number: the whole app is about **20,000 lines of code**. That's tiny for what it does. Enterprise apps often run into the millions.

8. How do updates get released?

We use a system called **Git**. Think of it as **track changes**, but for code.



Every time a developer saves an improvement, the app rebuilds automatically and the new version goes live. No downtime. Users just refresh the page.

If something breaks, we can **rewind time** — Git remembers every version and we can put an older one back within a minute.

9. What does it cost to run?

Right now, for the pilot: \$0 per month. Everything's on free tiers.

Once we roll out to lots of units, costs go up a little:

Stage	Monthly cost
Pilot (right now)	\$0
One region live	~\$60
All India live	~\$105
Full company worldwide	~\$400

For comparison: a **single manual audit** costs way more than that per month. And we do zero paper.

10. What could break, and what happens if it does?

Every part of the system has a plan for when it breaks.

If this breaks...	What happens	Fix
The screen (browser)	User sees an error message	Refresh the page
The brain (Render)	App goes down	Renders another copy — usually back in 30 seconds
The filing cabinet (Neon)	Nothing loads	Neon has automatic backups. We restore.
A specific feature	That page shows an error	Developer looks at logs, ships a fix
A photo we uploaded	Photo disappears	Real risk on the free tier — see below

One real limitation right now: on the free tier, photos uploaded during inspections might disappear if the app restarts (roughly once a day). The important data — who did what inspection when — is safe in the filing cabinet forever. Photos need a small upgrade for the enterprise rollout: about half a day of work.

11. What's next?

This is a **pilot**. It works today. It's live at <https://upl-fire-portal.onrender.com>.

For the enterprise rollout, we need three things:

1. **Executive sign-off** to make it the official system of record
2. **A tiny budget** (see cost table above) to move off the free tier
3. **Two people** to help roll it out to each unit — training + onboarding

Once those are done, we scale from **1 unit** → **all of UPL** in about 6 months. Full plan is in the executive deck ([docs/presentation/](#)).

12. Where do I look if I want to learn more?

I want to...	Read this
Understand the code in depth	DEVELOPER_GUIDE.md
Use the app as an inspector / admin	USER_MANUAL.md
Pitch this to my boss	presentation/SafetyVerse_Executive_Deck.pptx
Deploy it somewhere new	DEPLOYMENT.md
Just try it right now	Open https://upl-fire-portal.onrender.com and log in as <code>admin</code> / <code>Admin@123</code>

That's the whole system in one document. No hidden magic. No fancy words. Just a screen that talks to a brain that talks to a filing cabinet, wrapped in a nice orange colour and stuck on a QR code.